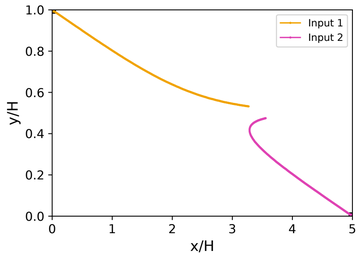
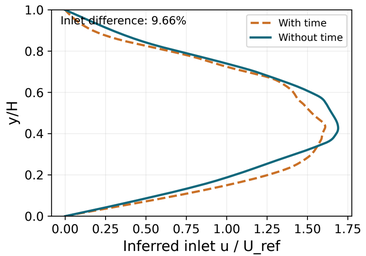
PI-CFO | Corner launches / experiment B | with time vs without measured time

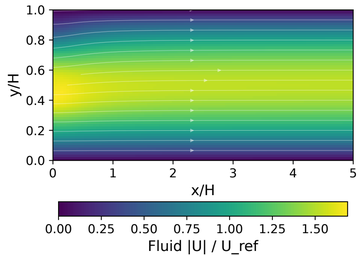
01 Given ordered paths



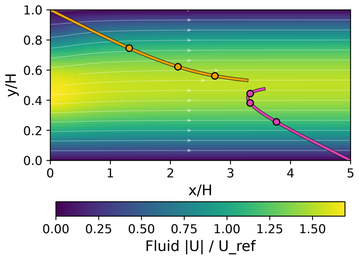
02 Two inferred inlets



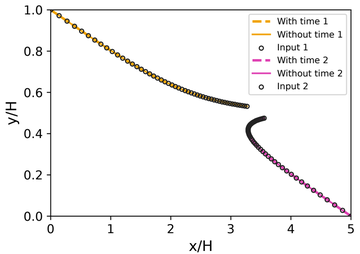
03 Flow without time



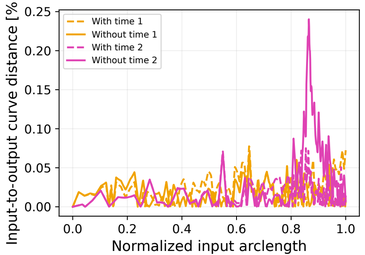
04 Particles in that flow



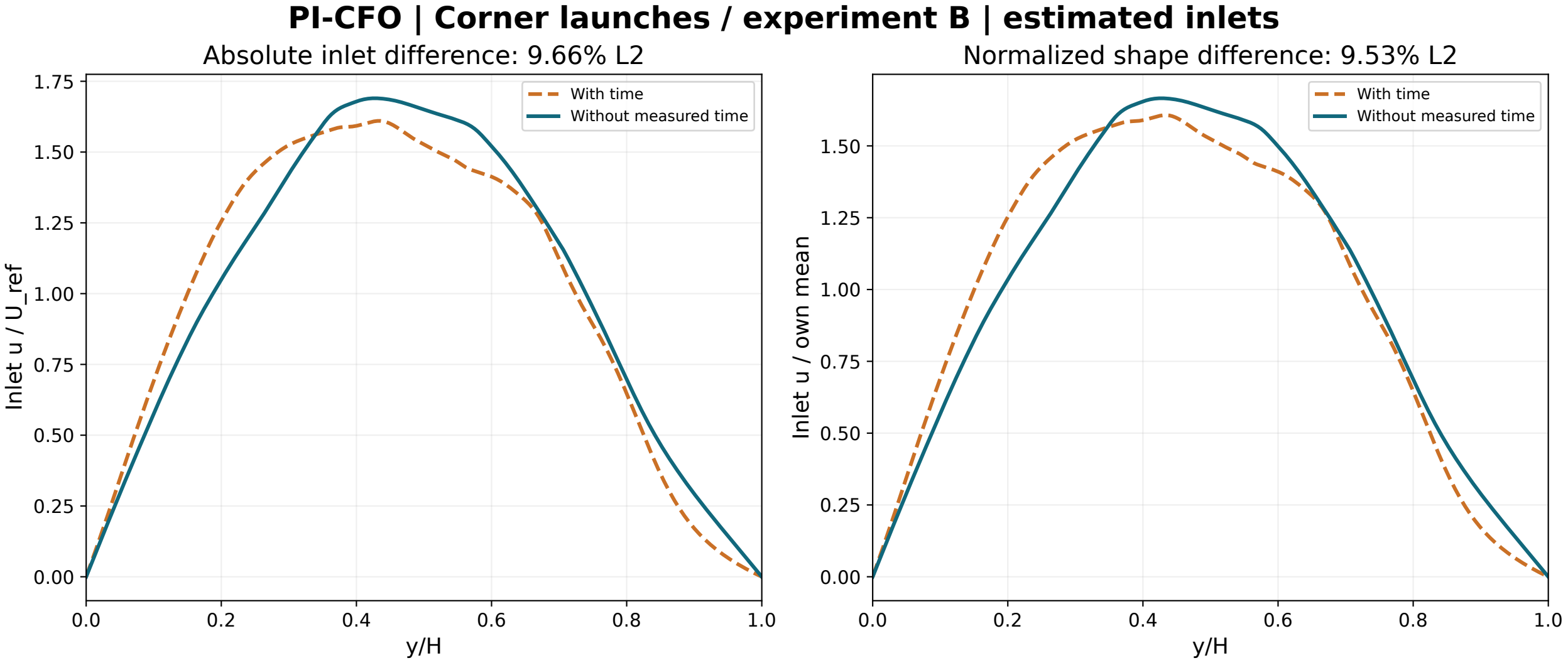
05 Both path predictions



06 Geometric mismatch

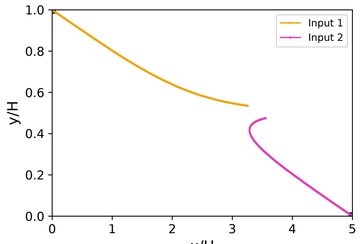


Solid: without measured time; dashed: with time; open circles: given geometry. Both outputs are independent ODE replays. Curve agreement does not guarantee matching inlet profiles.

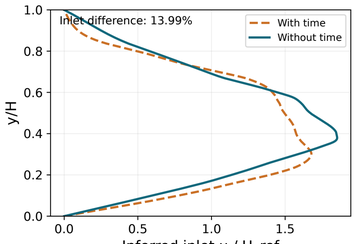

PI-CFO | Corner launches / experiment B | estimated inlets
Absolute inlet difference: 9.66% L2
Normalized shape difference: 9.53% L2
With time
Without measured time
Inlet u / U_ref
Inlet u / own mean
y/H
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
0.0
0.2
0.4
0.6
0.8
1.0

PI-CFO | Corner launches / experiment A | with time vs without measured time

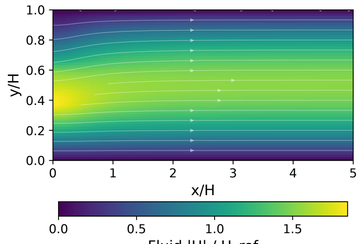
01 Given ordered paths



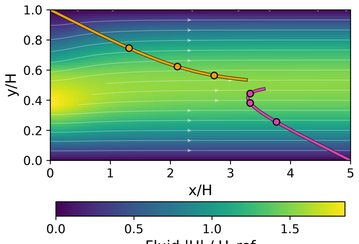
02 Two inferred inlets



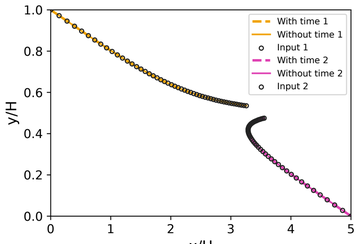
03 Flow without time



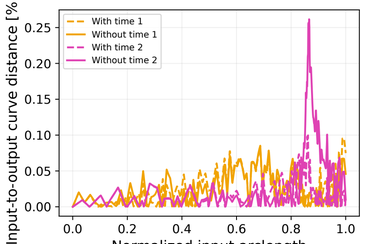
04 Particles in that flow



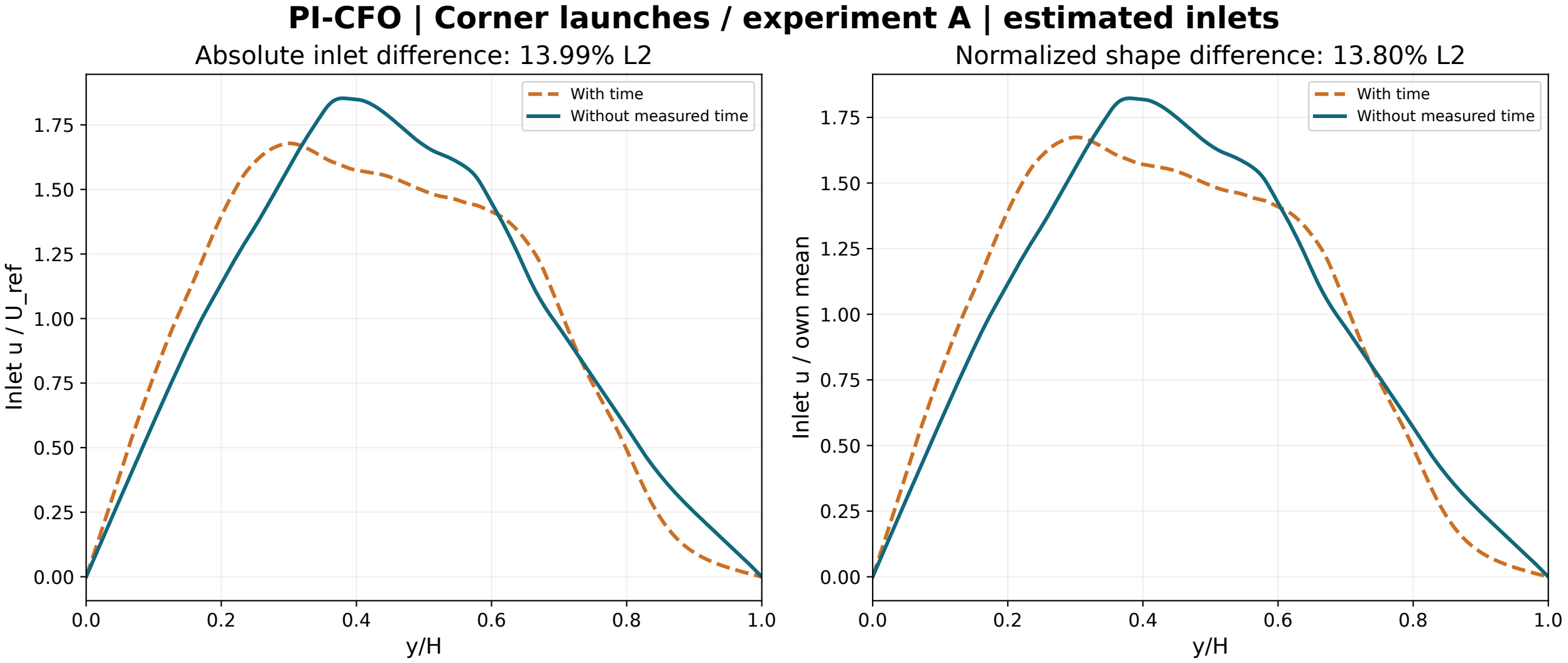
05 Both path predictions



06 Geometric mismatch

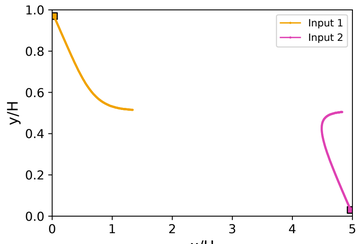


Solid: without measured time; dashed: with time; open circles: given geometry. Both outputs are independent ODE replays. Curve agreement does not guarantee matching inlet profiles.

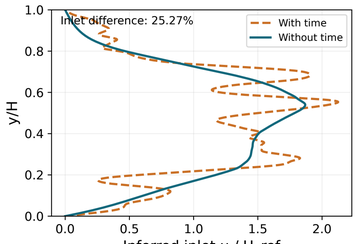

PI-CFO | Corner launches / experiment A | estimated inlets
Absolute inlet difference: 13.99% L2
Normalized shape difference: 13.80% L2
With time
Without measured time
Inlet u / U_ref
Inlet u / own mean
y/H
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
0.0
0.2
0.4
0.6
0.8
1.0

PI-CFO | Oblique launches | with time vs without measured time

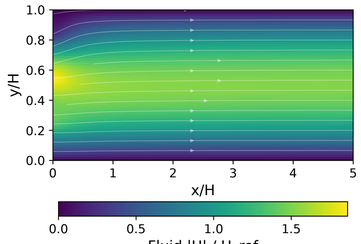
01 Given ordered paths



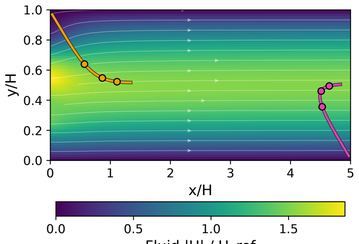
02 Two inferred inlets



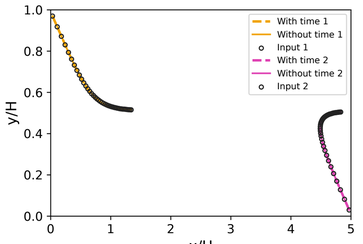
03 Flow without time



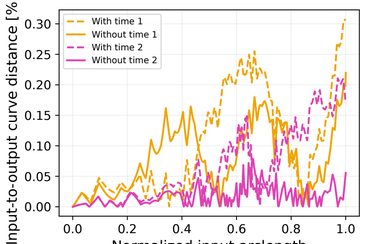
04 Particles in that flow



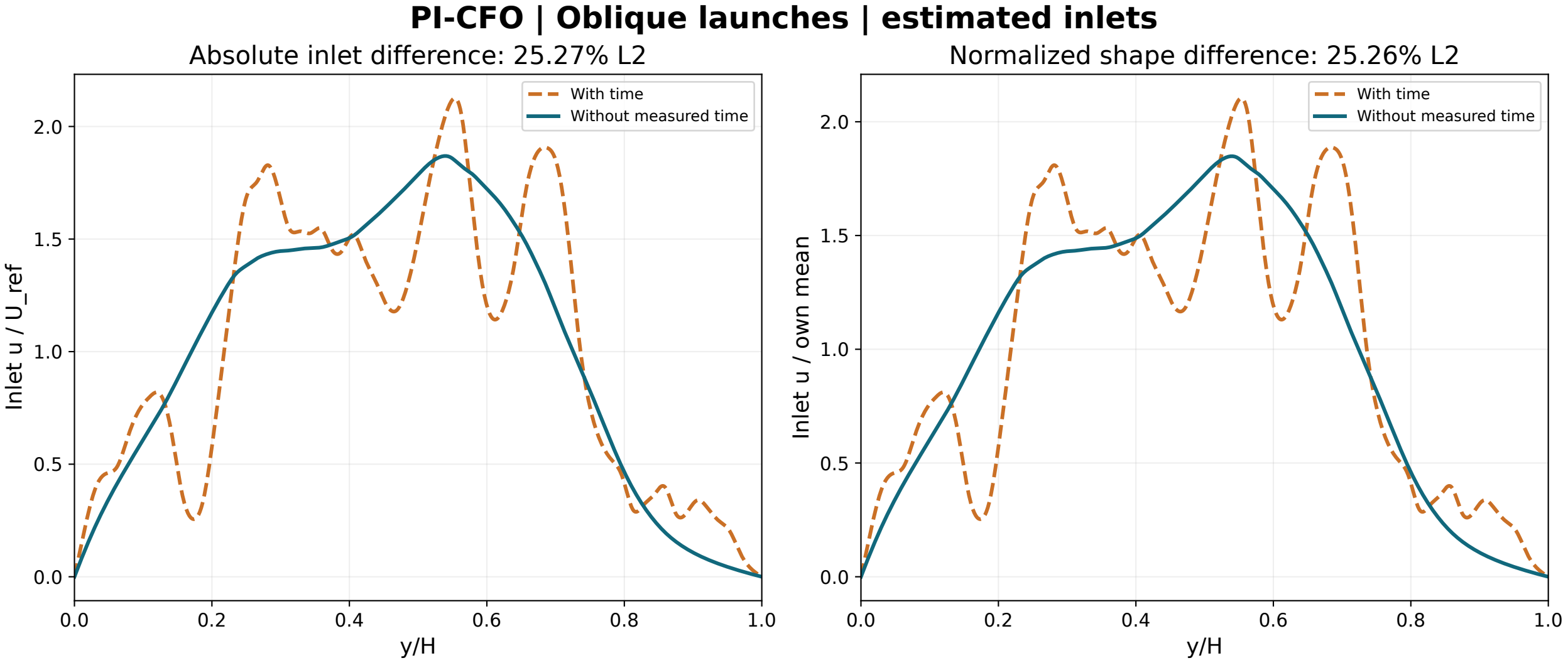
05 Both path predictions

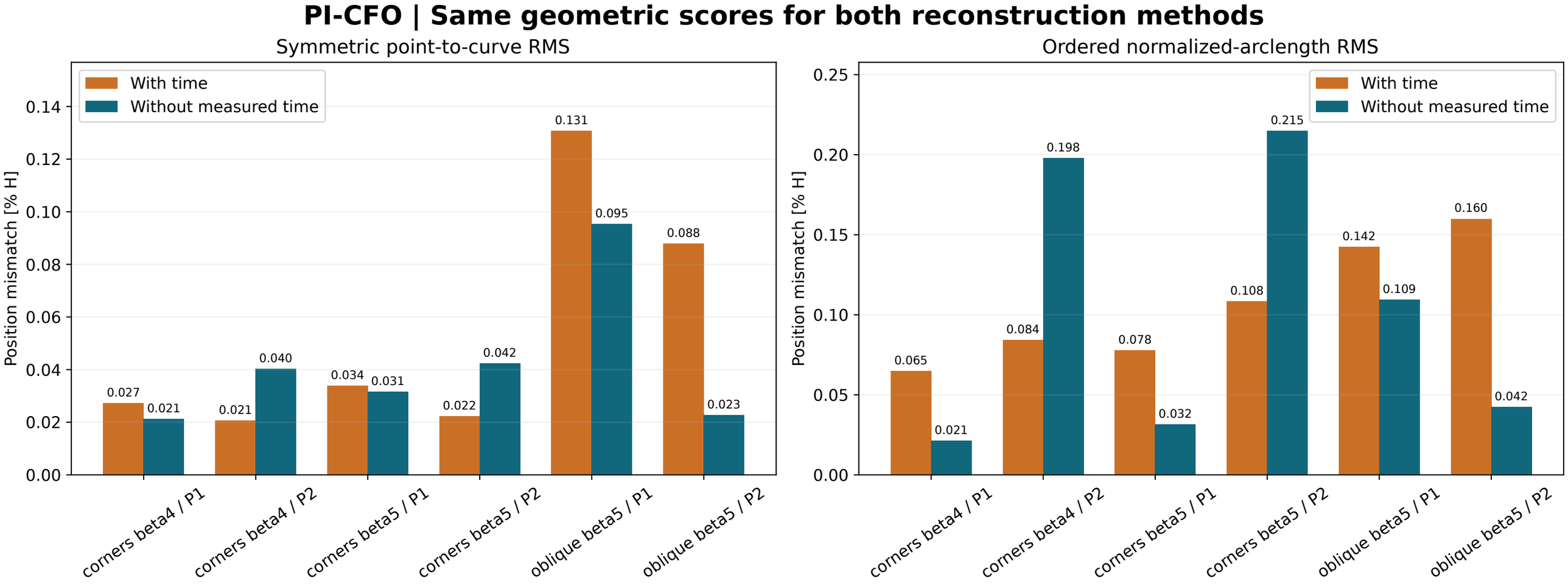


06 Geometric mismatch



Solid: without measured time; dashed: with time; open circles: given geometry. Both outputs are independent ODE replays. Curve agreement does not guarantee matching inlet profiles.


PI-CFO | Oblique launches | estimated inlets
Absolute inlet difference: 25.27% L2
Normalized shape difference: 25.26% L2
With time
Without measured time
Inlet u / U_ref
Inlet u / own mean
y/H
0.0
0.5
1.0
1.5
2.0
0.2
0.4
0.6
0.8


PI-CFO | Same geometric scores for both reconstruction methods
Symmetric point-to-curve RMS
Ordered normalized-arclength RMS
With time
Without measured time
Position mismatch [% H]
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.05
0.15
0.20
0.25
0.027
0.021
0.021
0.040
0.034
0.031
0.022
0.042
0.131
0.095
0.088
0.023
0.065
0.021
0.084
0.198
0.078
0.032
0.108
0.215
0.142
0.109
0.160
0.042
corners beta4 / P1
corners beta4 / P2
corners beta5 / P1
corners beta5 / P2
oblique beta5 / P1
oblique beta5 / P2

Similar paths need not imply identical inlet profiles

